t (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

ANNUAL REVIEWS

Annual Review of Sociology

The Knowledge-Based Economy and the Global South

Nitsan Chorev and Amanda C. Ball

Department of Sociology, Brown University, Providence, Rhode Island, USA;
email: nitsan_chorev@brown.edu, amanda_ball@brown.edu

Annu. Rev. Sociol. 2022. 48:171–91

First published as a Review in Advance on April 19, 2022

The *Annual Review of Sociology* is online at soc.annualreviews.org

https://doi.org/10.1146/annurev-soc-080321-071214

Copyright © 2022 by Annual Reviews.
All rights reserved

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Keywords

knowledge-based economy, innovation, industrialization, upgrading, Global South, low-income countries

Abstract

Research on knowledge-based economies and innovation tends to focus on the highest-ranked knowledge-advanced countries in the Global North and on high-growth countries in the Global South. In this article, we review the research on knowledge-based economies in the Global South and identify two important analytical shortcomings: the tendency to conflate upgrading with innovation and the tendency to focus on domestic factors. Moving beyond the familiar case studies, we explore the literature on pockets of knowledge-based industries in economies that are not dominantly knowledge-based. We then shift the focus from the private sector to explore the literature on state-led knowledge-based innovation, particularly welfare provisions and surveillance. We conclude by challenging the notion that innovation is simply the next step in industrial development and call for a more historically and geographically specific sociological approach to understanding innovation and knowledge-based industries, particularly in the Global South.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

1. INTRODUCTION: WHAT IS A KNOWLEDGE-BASED ECONOMY?

In *The Poverty of Philosophy*, Karl Marx (1847, p. 119) famously stated that "the windmill gives you society with the feudal lord; the steam-mill, society with the industrial capitalist." Setting the question of technological determinism aside, Marx here clearly sees technology—and therefore knowledge—as key for political-economic transitions. Capitalism's distinctiveness may therefore be that of continuous technological innovation as part of the same economic order. This continuity is captured in Joseph Schumpeter's (1942, pp. 82–83) "creative destruction"—the process of "industrial mutation that continuously revolutionizes the economic structure from within, incessantly destroying the old one, incessantly creating a new one."

Political economists have long tried to capture the underlying logic of capitalism's revolutionizing from within (Bell 1973, Krippner 2012, Zuboff 2019), and the centrality of knowledge to that process has long been recognized (Muzaka 2019). Already in 1890, Alfred Marshall realized that "knowledge is our most powerful engine of production" (Marshall 1890, p. 115; see also Schumpeter 1911). Knowledge's positive growth effect relies on a unique quality—unlike land, capital, or labor, ideas and technological advances are nonrivalrous, so that consumption by one does not impede its simultaneous consumption by others (Romer 1990, 1993; Jones 1999). Nonetheless, some argue that in the postindustrial economy, knowledge has become the only economic resource that matters (Drucker 1993, Cooke & Leydesdorff 2006, Collinge & Staines 2009) and is the basis of economic power (Bell 1973, Burton-Jones 2003). Only the postindustrial economy is truly a knowledge-based economy.

Are we correct to define the current phase of capitalism as uniquely knowledge-based, given that knowledge has always been an aspect of production? Not even Frederick W. Taylor's attempt to separate conception from execution or Henry Ford's assembly line could do away with workers' either formal or tacit knowledge. Although we agree that knowledge is not distinctive to our era (Collinge & Staines 2009), we contend that it is still useful to consider knowledge key for today's capitalism in particular, not only because the importance of knowledge has been accelerating (OECD 1996) but also because knowledge is utilized in novel ways. Today, the focus is on knowledge about knowledge, with important multiplying effects. One example is the "expansion of ICT [information and communication technology] and the Internet, which has dramatically enhanced our ability to access, exchange and store knowledge" (Collinge & Staines 2009, p. 166). A World Bank (2007, p. 2) report offers other examples:

> An increase in computer speed and processing power has facilitated other technological leaps. Developments in genetics and associated biotechnologies have spurred tremendous progress in the life sciences and their applications. Advances in the manipulation of matter have led to the invention of new materials and nanotechnologies.

Because resources devoted to innovation are expected to enable output to grow faster than the growth of capital and labor inputs, many argue that the success of enterprises and national economies depends on generating rather than merely utilizing knowledge (Lundvall 1992, Steinmueller 2002, Collinge & Staines 2009). The production of knowledge—that is, innovation—is seen as a necessary condition for continuous growth and the most fundamental aspect of the knowledge-based economy (Acemoglu & Akcigit 2012). Even more narrowly, some see a knowledge-based economy as "the exploitation of new knowledge in order to create more new knowledge" (Cader 2008, p. 119).

Various methods have been adopted to measure knowledge-based economies and rank countries according to their knowledge-based capabilities.[1] Common measures rely on (*a*) knowledge

[1]These include the World Bank's Knowledge Assessment Methodology (KAM) database, the UNCTAD Innovation Capability Index (**https://unctad.org/en/PublicationsLibrary/dtlstict2013d8_en.pdf**), and the

use, such as the use of advanced machines (McGranahan & Beale 2002) or of electronic communication and exchange (Cader 2008); (*b*) knowledge-related input indicators, such as research and development (R&D) spending (OECD 1996); (*c*) knowledge-based output indicators, such as patents or scientific publications (Machlup 1962, OECD 1996, Cader 2008); and (*d*) knowledge labor (Eliasson et al. 1991, Beck 1992, Burton-Jones 1999, Wojan 2000, Cader 2008) that measures the availability of knowledge workers "with high degrees of expertise, education or experience," where "the primary purpose of their job involves the creation, distribution, or application of knowledge" (Davenport 2005, p. 10; see also Powell & Snellman 2004, Cader 2008).[2] Note that some of the indicators measure knowledge-based use or upgrading while others measure innovation capabilities, an issue we return to below.

Based on these measures, many of the highest-ranked knowledge-advanced countries are high-income countries in the Global North. Sociologists follow suit and focus their research on a limited geographical scope. Economic sociologists focus almost solely on countries in the Global North (e.g., DiMaggio 2001). Sociologists of development, in turn, study countries "closely connected to the advanced core of the world economy" (Saxenian & Sabel 2008, p. 379), such as Ireland, Finland, Iceland, and Israel, as well as South Korea and Taiwan. In Section 2, we review the literature that looks at high-growth countries in the Global South. We identify two important limits to that literature. First, it tends to explain upgrading and innovation by referring to domestic conditions. We argue for the need to consider transnational factors as well—including global economic and geopolitical as well as global-historical aspects. Second, the literature tends to conflate the analysis of innovation with the analysis of upgrading. Although both the use of knowledge-based technologies (developed elsewhere) and the production of knowledge fall under the label of "knowledge-based," we maintain that different tools are required to explain innovation compared with earlier stages of industrialization, including the stage of knowledge-based upgrading.

In Section 3, we move away from the mainstream literature by considering pockets of knowledge-based industries in economies that are not dominantly knowledge-based. We argue that looking at innovation in contexts of scarce knowledge resources offers new insights beyond those emerging out of higher-ranked knowledge economies. As importantly, looking at innovation where it is not expected offers a critical perspective of technology absorption—forcing us to consider the possibility of dependence, inappropriateness, appropriation, and exploitation.

Knowledge-based economies include not only manufacturing but also services, and much scholarly attention is being paid to the digital sector—namely online platforms (e.g., Google, Facebook, Alibaba) and platform-enabled services (e.g., Airbnb, Uber) (IMF 2018, Zuboff 2019). But the digital services also shape sectors such as education, healthcare, and social work, which are often under the domain of the state. In Section 4, we advance beyond the private sector to explore state-led knowledge-based practices. We discuss two state-led domains of knowledge-based innovation—welfare provision and surveillance—and call for a broader future research agenda. We conclude our review by suggesting the need for a more inclusive and historically specific approach to studying knowledge-based economies and innovation that will extend and strengthen current approaches.

2. KNOWLEDGE-BASED INNOVATION AND USE IN HIGH-GROWTH COUNTRIES IN THE GLOBAL SOUTH

Innovation capabilities have been concentrated in the United States, Western European countries, and Japan (Altenburg et al. 2008, Rikap 2019, World Bank 2019; see also Heidenreich 2012).

Global Competitiveness Index (**http://www3.weforum.org/docs/WEF_TheGlobalCompetitivenessReport2019.pdf**).

[2]Measures can be selected strategically to manipulate the level of success (Wong 2011).

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

High-ranked innovation countries have been able to protect their dominance over the production of knowledge by various means, including international agreements. The 1994 Agreement on Trade-Related Aspects of Intellectual Property Rights, which was negotiated in the course of establishing the World Trade Organization (WTO), required all WTO members to adopt and enforce high minimum standards of intellectual property protection. Countries with loose intellectual property laws now had to enforce more stringent patent laws, which helped prevent the diffusion of technologies and slowed the upgrading and innovation in lower-ranked knowledge countries (Chang 2002).

Still, the technological gap between the core and the periphery is narrowing. China, India, and Brazil have declared their intention of becoming global knowledge powers, and countries like Mauritius are trying to transform into knowledge hubs (Muzaka 2019). Such announcements are often considered empty gestures, but Ford (2017, p. 57) shows that in the United States, intellectual property laws contributed to industrialization through the creation of a particular political culture "in which collective hopes for the future are placed in new technologies." Indeed, against expectations, China has developed innovation capabilities in goods-based industries, but also in ICT-led fields such as artificial intelligence (AI) and machine learning. Companies like Alibaba, Baidu, Tencent, and TikTok, which dominate Internet browsing, e-commerce, messaging, gaming, and other consumer Internet products in China, are extremely successful innovation companies (Wade & Shan 2019). India developed capabilities in the information technology (IT) and business process outsourcing sectors (Gereffi & Ong 2006). In Argentina, companies export architectural plans and legal and accounting consultancy work, and are involved in software and mobile app development, web design, and film and video game production (Opertti 2018). Given the technological, political-economic, and legal obstacles, how do scholars explain the ability of these countries to innovate?

In identifying what is conducive to knowledge-based upgrading or innovation,[3] most scholars emphasize domestic conditions. Local institutions that are considered beneficial for innovation and technological upgrading include "competitive democracy, decentralized government, property rights, free markets, [and] different styles of capitalism" (Taylor 2012, pp. 121–22). Other conditions include "wide availability of a cheap and malleable workforce," access to bank loans, and "huge domestic markets" (Reslinger 2013, p. 389). Although these conditions are often the same as those recognized to be essential for industrialization in general, scholars also identify a few factors that are unique for knowledge-based development. One such condition is knowledge infrastructure, including public education and research universities (Taylor 2016). Scholars, in turn, identify variation in that infrastructure. Gereffi & Ong (2006) find that India relies on private colleges while in China the government is central for promoting engineering education and nanotechnology research (see also Liu et al. 2018).

Of all the proximate domestic factors leading to technological upgrading and innovation, the one that receives the most attention is the state. This is somewhat surprising. Of course, state-led industrial policies have long been recognized as instrumental for late development (Amsden 1989, Evans 1995). However, a neoliberal attack on the interventionist capacity of the state (Taniguchi & Babb 2009) combined with the decentralization of production (Chorev 2007, Breznitz & Murphree 2011) called into question the ability of the state to decide "what, when and how much

[3]As we discuss below, the relevant literature does not always differentiate among learning, upgrading, and knowledge-based upgrading, and it does not consistently differentiate between knowledge-based upgrading and innovation. R&D capabilities are at times considered an aspect of upgrading, for example. For this reason, we discuss (knowledge-based) upgrading and innovation together.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

to produce" (Amsden 1989, pp. 143–44). In addition, the perception that the source of growth is no longer (low-tech) production but (knowledge-based) innovation implies that lead firms of global value chains, rather than the state or other local actors, control the decisions of local firms, again undermining the state's capacity to guide the market (Ó Riain 2004, Yeung 2014, Thurbon & Weiss 2019).

Yet, scholars argue that state actions continue to benefit industrial upgrading and innovation (Reslinger 2013, Perez-Aleman & Alves 2017). Some countries still employ old types of protectionist policies to develop new industries. China, for example, is leading the way in digital protectionism. By blocking Facebook and Google, the government not only enabled the rise of domestic platforms like WeChat and Weibo but also gave them sole access to the vast troves of data collected (Ding 2018). The type and impact of interventions have changed enough, however, that scholars now refer to neo-developmental, developmental network, entrepreneurial, or flexible states. Policies, for example, center on targeted government investment in R&D and training, building of facilities and science parks, support for marketing and exporting, and patent subsidy programs (Li 2012, Taylor 2016, Maggor 2020).

One central role of the state is coordinating networks and providing brokerage services. Negoita & Block (2012) show how Chilean government agencies helped in creating globally competitive firms in agro-industries through the recruitment of entrepreneurs, connecting these entrepreneurs to technologies they needed, helping them build effective networks that included financial support, and working with them on an ongoing basis to solve problems. Kim (2019, p. 160) similarly shows how Taiwan and South Korea expanded their green energy sectors by establishing "hybridized industrial ecosystems" of public-private partnerships. Others find similar instances of government-business collaborations, including with transnational capital (Dent 2003, Lee et al. 2014, Thurbon & Weiss 2019).

Novel institutional capabilities have also been found useful for knowledge-based developmental outcomes (Kim 2012). Institutions that could facilitate industrialization in the past have arguably been overembedded and overcentralized for the new economy (Ó Riain 2004). Block (2008, p. 172) explains why: "The main focus of the [state] is to help firms develop product and process innovations that do not yet exist, such as new software applications, new biotech medications, or new medical instruments. For this, the older. . .model is basically irrelevant because there is no international leader that firms can imitate." Instead of highly coordinated state policies, China's rapid economic growth suggests the effectiveness of trial-and-error economic experimentation that is led by decentralized entities (Breznitz & Murphree 2011, Breznitz et al. 2018). Others, however, see less of a contrast between former and current state arrangements. On the basis of Korea's telecommunications sector, Kim (2012) concludes that neo-developmental institutions are more of a continuation of than a replacement for older institutions.

The contemporary literature on the state, by emphasizing networks, converges both with the literature on state-society relations and with market-centered approaches to development (e.g., Tang & Hussler 2011; but see Nee & Opper 2012). From scholars of state-society relations we learn that effectiveness of state support may depend on the traits of the private networks (Samford 2017, Fuentes 2019), as well as on the density and quality of the relationships between supporting state institutions and the enterprises (Altenburg et al. 2008). Although scholars identify indigenous sources of learning (McDermott et al. 2009), it is more common for later industrializers to learn from others (Amsden 1989), and much of the literature focuses on networks that are part of "global innovation chains," where the generation of innovation is elsewhere [Archibugi & Iammarino 2002, p. 99; on multinational companies (MNCs), see Heidenreich 2012].

These international networks include both inward foreign direct investment (IFDI), where MNCs invest in local enterprises, and outward foreign direct investment (OFDI), where local

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

enterprises venture internationally. In regard to IFDI, scholars have long debated the extent to which foreign investment leads to the diffusion of technology (Prasad 1998, Larraín et al. 2000, Zhang & Duysters 2009, Demena & van Bergeijk 2019), and IFDI's contribution to innovation is similarly contested. While some see integration into global production networks as a sure path for upgrading (Reslinger 2013), others warn of "shallow integration" (Chen 2014, p. 226). Counterintuitively, Chen (2014) shows that IFDI by smaller companies is more conducive to learning than that by larger companies, due to a narrower technological gap. Even a shallow mode of market integration may affect domestic institutions, however (Langbein & Markiewicz 2019). In turn, large companies in the Global South actively engage in OFDI. One of the unique features of the so-called new multinational enterprises is that their international expansion brings them to developing and developed countries simultaneously. They enter markets of developing countries to gain size and operational experience and to generate profits; they enter developed countries not because they already have the necessary capabilities to compete in those markets, but to build capabilities in the first place (Guillén & García-Canal 2009; see also Mathews 2006, Aulakh 2007, Duysters et al. 2009).[4]

Some argue that national innovation rates depend on both domestic and transnational networks (Taylor 2016), showing that although innovation certainly depends on domestic networks between the science and technology labor force, local entrepreneurs, and investors, it also requires international linkages between domestic innovators and global markets for exports, capital, and sources of knowledge and skills. Zhang & Duysters (2009) find that IFDI impacts indigenous firms' performance only when strategic alliances have been established beforehand. However, integration into international networks through investment is not the only way to learn and upgrade. To this end, Mehri (2015) shows that the automobile industry in Iran benefited from foreign engineering consulting firms.

In China, the strategies leading to the unprecedented success of Huawei—the company is the largest telecom equipment manufacturer in the world, the second largest producer of smartphones, and the only company that can already produce at scale and cost all the elements of a 5G network (Zhang & Duysters 2009, Kaska et al. 2019)—demonstrate the importance of both domestic and international networks, as well as state support. Huawei's first technological stage-skipping in the telecommunication sector was the result of collaboration with and access to the technology of Shanghai Bell, which was the first Sino-foreign joint venture in China. Over the years, Huawei established other technological alliances with Japanese, American, and European enterprises, including Siemens and Motorola, and with management consultant companies, and it established R&D centers outside China. State support was also crucial to Huawei's success. Early on, to avoid direct competition with Shanghai Bell, Huawei expanded only in the rural market and small cities, with political support from the Chinese government. Huawei also received financial credits for R&D funding (Mu & Lee 2005, Zhang & Duysters 2009, Joo et al. 2016).

Learning may also be the outcome of migration. In the early nineteenth century, the poaching of skilled workers allowed Germany to adopt technologies developed in Britain (Chang 2002), while industrial development in Latin America benefited from the immigration of entrepreneurs and scientists from the United States and Europe (e.g., Gereffi 1978). Similarly, Saxenian's (2006) study of software clusters in India and China shows how movement of skilled personnel could change the geographical division of innovation. Saxenian (2006) emphasizes, in particular, the

[4]While learning by investment may be an intentional strategy of new multinational enterprises, it has always been part of the experience. Kwon (2005) describes how Japanese manufacturers of cars and electronics adopted their model of production after seeing how Americans had amended the original Japanese model.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

role of return migrants, whom she calls the "new Argonauts"—technically skilled entrepreneurs, engineers, and scientists, with substantial experience in the United States, who apply their skills and capital in their countries of origin. The most significant contribution of these professionals is not necessarily the direct transfer of knowledge, but participation in external search and domestic institutional reform (Altenburg et al. 2008, Saxenian & Sabel 2008). This return migration can be market-driven, or it can be sponsored by the government (O'Meara 2019).

These analyses offer valuable insights but also leave important questions unanswered. In regard to the role of the state, for example, studies are able to show that interventions took place but less able to show whether or how these interventions lead to success. And what does the state get in return? In analyzing state-capital relations, few scholars followed Amsden (1989) or Chibber (2003) and asked how is the state able to discipline capital or whether the state is able to ensure that "the. . .'spillovers' produced by the state's investment in innovation are not appropriated by private actors" (Maggor 2020, p. 452). In addition, only a few studies examine how noneconomic considerations, such as environmental or public health concerns, guide state investment and enhance or inhibit, for example, catching up in green technology or innovation in pharmaceuticals (Schüren 2013).

More fundamentally, we take issue with the lack of analytical differentiation in most of the relevant literature between upgrading and innovation of knowledge-based technology. This may stem from the assumption that knowledge-based innovation is simply another stage in industrial development. Block & Negoita (2016, p. 59), for example, explicitly state that "the requisites for effective developmental state initiatives are now similar whether the goal is catching up or indigenous innovation." This linear logic allows scholars to apply the explanations used for industrialization for innovation as well, with only modest modifications (somewhat different tasks for the state, somewhat different types of networks, etc.). In contrast, we suggest that innovation requires a departure from the linear logic of industrialization and catching up, for a number of reasons.

First, there are good theoretical reasons to question the equivalence made between upgrading by learning and innovation. While upgrading is about following a path already taken by others, with innovation, the path forward is unknown. Among other implications, this has a major effect on the ability of the state to direct the market. It is difficult for the state to pick a potential winner when there is no clear or predetermined path to success, as is necessarily the case with innovation. Second, learning (for either industrialization or upgrading) and innovation require different sets of skills. Indeed, Amsden's (1989) insight that technological capabilities can be learned from others and therefore do not have to be developed indigenously exactly implies that learning can bypass the need for innovation and, by extension, that learning and innovation require different processes. Third, the literature assumes a sequential process in which the initial stage of catching up later enables innovation, but we know that latecomers may skip certain stages or create their own path that is different from those of the forerunners (Lee & Lim 2001, Gereffi & Ong 2006, Lee 2013, Joo et al. 2016). Some argue that a latecomer's catch-up is possible only by leapfrogging, that is, taking an industrialization shortcut or detour (Appelbaum et al. 2011, Lee 2019). Fourth, the literature implies that all innovations require the same conditions, but not all innovations are the same. Consider, for example, the difference between purely novel technologies and second-generation innovation that includes the mixing of established technologies and products, which was the strategy employed by both Baidu, the dominant search engine in China, and TikTok (Breznitz & Murphree 2011, Akenzua 2019). Different types of innovation may reasonably require different sets of conditions. Finally, the literature has mostly neglected to consider failed cases. For example, in South Korea, Taiwan, and Singapore, investments in biotechnology have performed quite badly (Wong 2011), which confirms our suggestion that innovation does not follow upgrading. In short, innovation is sufficiently different from upgrading that we cannot assume that the

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

conditions leading to industrialization and upgrading are the same as the conditions permitting innovation.

Innovation, then, is different than previous stages of industrialization. In addition, today's political-economic context—the context under which innovation is supposed to take place—is different from previous contexts, an issue that the literature has not paid much attention to. In the literature on innovation, like much of the literature on industrialization that preceded it, the state is reduced to its economic or other government functions and is judged by its bureaucratic capacity, making the state strangely apolitical (but see Chen 2017). Yet, through even minimal interventions, the state still selects potential beneficiaries and designates losers, and interest groups compete over the distributional outcomes of scientific and technological changes (Taylor 2016). In our view, the scholarship on upgrading and innovation has not done enough when it comes to opening the black box of the relations between state and capital. For example, in discussions on development, the electric vehicle industry is analyzed as an opportunity for leapfrogging (Goldemberg 1998, Pérez 2001), but Hou & Li (2020, p. 66) remind us that many of the challenges are political, since electric vehicles have "a strong and well-established competing interest—the gasoline auto industry."

The scholarship has paid even less attention to the black box of foreign policies and geopolitics, even though the relations of local companies with foreign capital are directly influenced by the global political-economic context. Geopolitics is clearly relevant to the establishment of global commodity and innovation chains. Geopolitics may also be relevant to a state's industrial priorities. Taylor (2016) suggests, for example, that only political elites who feel threatened by external hazards more than by domestic rivals take the risk of investing in innovation. Another underexplored aspect of the geopolitics of learning is the violation of norms of international commercial exchange, including through reverse engineering, poaching, or industrial espionage (Amsden 1989, Chang 2002). International rules against such practices—by strengthening intellectual property rights, for example—are one of the ways developed countries attempt to inhibit learning and innovation elsewhere (Chang 2002). Global political-economic analysis might shed light on the fact that Huawei, which as the only capable supplier of 5G technology today is a pioneer in what might bring a fourth industrial revolution, is at the center of fraud and intellectual property theft investigations in the United States (Ding 2018, Halpern 2019, Kaska et al. 2019). In the case of Huawei and other Chinese communications and video surveillance technology manufacturers, geopolitics moves beyond international economic competition to include issues of cybersecurity and fear of Chinese surveillance, especially because the Chinese government has access to data collected by Chinese companies (Kaska et al. 2019). Although some countries banned TikTok, the video-sharing social networking service, for allegedly encouraging pornography and other illicit content, the fear of data ending up in the hands of the Chinese government is often part of the motivation (Davis 2019).

Finally, we need to consider global historical conditions that affect success and failure. As Chang (2002) and Weiss (2005) show, rich nations have put in place a system that constrains opportunities for developing countries to rise through the global hierarchy. However, the literature is mostly concerned with global trade rules and market structures that affect the extent to which states are allowed to support local industries. But other local conditions, such as state capacity or available knowledge infrastructures, are proximate causes that require further explanations—why, for example, do some countries have better research institutions than others? Past interventions shape contemporary conditions—for example, the postliberalization success of the IT and pharmaceutical sectors in India is due to state policies introduced back in the 1970s (Chaudhuri 2005, Saraswati 2008). In turn, to understand India both in the 1970s and today, we need to look at colonial and precolonial factors as well. Such explorations will give us a better understanding of

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

knowledge-based economies, as well as of pockets of knowledge-based industries, to which we refer next.

3. EXCLUSION AND POCKETS OF INNOVATION IN LOWER-RANKED KNOWLEDGE ECONOMIES

The insight that growth is driven by innovation is compatible with the argument that the so-called Great Divergence, whereby poor countries are falling further behind rich ones, is due to differences in technology adoption (Pritchett 1997; Comin & Mestieri 2018, p. 139). Still, zones of innovation exist also in countries that "lack most of the institutions that economists view as preconditions for growth" (Saxenian & Sabel 2008, p. 380). Scholars have observed the "emergence of clusters" of innovation (Saxenian & Sabel 2008, p. 380) in countries like India and China but have shown less interest in knowledge-based pockets in other low- or middle-income countries. Scholars also ignore the cause of technological exclusion in the first place.

As for the cause of exclusion, the global unevenness of knowledge production and use is only a recent manifestation of a broader pattern of marginalization and an outcome of legacies of exploitation and appropriation. Under colonialism, Western technological innovation has threatened production elsewhere through the creation of alternative, cheaper products (Headrick 1988), at a time that colonial administrators discouraged skill-building among indigenous populations. Even basic innovations, such as electricity, the internal combustion engine, the telegraph, and telephones, have spread only unevenly across continents (Kleine & Unwin 2009). By the time that, with globalization, technology was becoming more easily diffused, adopting it did not necessarily bring economic growth. Labor-saving technological progress brought about "premature deindustrialization," which "blocks off the main avenue of rapid economic convergence in low-income settings" (Rodrik 2016, p. 28). Knowledge amplified the problem:

> Many services, such as IT and finance. . .could play the escalator role that manufacturing has traditionally played. However, these service industries are typically highly skill-intensive, and do not have the capacity to absorb—as manufacturing did—the type of labor that low- and middle-income economies have in abundance. (Rodrik 2016, p. 29)

This argument highlights the paradox of needing access to knowledge to take advantage of knowledge (see also Whittaker et al. 2010). Ewusi-Mensah (2012) finds that in the absence of knowledge infrastructure, the relationship between IT investments and firm productivity is weak, which leads to less investment in sub-Saharan Africa compared with other countries in the same economic development stage (see also Bollou & Ngwenyama 2008).

Systemic exclusion and appropriation notwithstanding, pockets of knowledge production do exist in lower-ranked knowledge economy countries. These pockets are often made to be invisible, however. Invisibility is at times an outcome of the fact that the types of innovation found in middle- and low-income countries are second-generation innovation (Breznitz & Murphree 2011) or incremental innovation, that is, marginal and continuous adjustments to existing products or processes (Shadlen 2011, Tang et al. 2016, Fransen & Helmsing 2017). Innovation in low- and middle-income countries is made invisible also because "being local carries a stigma" (Takhteyev 2012, p. 7). This may be why consumers do not know that the programming language Lua, used extensively in the video games World of Warcraft and Angry Birds, was developed in Brazil: "Local developers are thus themselves involved in replicating the asymmetries from which they suffer, as a possible necessary condition for their success" (Takhteyev 2012, p. 7). In other cases, such as the movie industry in Nigeria (Miller 2016, Christensen et al. 2019) or the technology industry in Ghana (Beerepoot & Keijser 2015), invisibility is a product of expanding to regional rather than international markets.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

In discussions on pockets of knowledge-based development in lower-ranked knowledge economies, the contributing factors identified are often similar to those mentioned in regard to higher-ranked knowledge-based economies, with a similar emphasis on learning through local networks and international partnerships (McDermott et al. 2009, Ndemo & Weiss 2017). Yet the study of lower-ranked knowledge economies highlights factors that are obscure elsewhere. For example, Silicon Savannah in Nairobi shows that location matters—clusters may depend on how attractive cities are for the creative class (Rosenberg & Brent 2020). As for the role of the state, the Nigerian film industry reveals the presence of informal substitutes that replace effective government support, such as the use of guilds instead of courts in solving labor disputes (Miller 2016; see also Bramann 2016). Similarly, the celebrated case of M-Pesa in Kenya, a mobile phone–based money transfer service, suggests that innovation can also happen in a shifting bureaucratic and patronage-based political context (Tyce 2020).

In turn, irrigation projects in Tanzania expose the blurry boundaries between the public and the private that are made too rigid in the literature on the developmental network state (Harrison & Mdee 2018). As another example, M-Pesa was developed by a public-private partnership that includes Safaricom, a Kenyan mobile network provider that is partly government-owned (Park & Donovan 2016), but the project received financial support from the UK Department for International Development. Foreign aid, in this and other cases, serves as a channel of technology transfer and enables innovation in contexts where it is difficult to obtain technical know-how through private investments (Chorev 2020).

The study of lower-ranked knowledge economies also highlights the risks and costs of borrowing technology from elsewhere beyond what we have already discussed above. One risk is technological dependence. Wade (2002, p. 443) warns that "as developing countries participate in ICTs, they become more vulnerable. . .to the quasi-monopolistic power of providers of key ICT services." A related risk is dependence on inappropriate technology (Stewart 1977). In low-income countries, infrastructural barriers may inhibit and potentially distort the use of imported knowledge. However, there is also the possibility of adaptation, as when Uber introduced cash payments in Ghana to make its services more suitable (Simons 2019). The use of appropriate technology may carry its own barriers. Because M-Pesa uses flip phones rather than smartphones, its services are accessible to a greater population in Kenya, but, by relying on older technologies, it arguably reproduces the technological gap between Kenya and rich countries.

Another cost is the possibility of appropriation, or what Benjamin (2009, p. 343) calls "upstream inclusion and downstream exclusion." Governments may fight back. In the case of extraction of genomic data, Benjamin (2009) examines how governments in some developing countries try to own their citizens' DNA—which has become a natural resource—to develop their own genomic information. Hicks (2019) describes governments' attempts to prevent "digital colonialism" by making foreign companies build data centers locally. Such data centers may boost digital industrialization; they may also increase governments' access to personal information. Ownership may not prevent dependence, however. In Mexico, genomic labs are in part owned by foreign investors. This extraction of genomic or online resources merely complements the extraction of traditional material resources, which is still going on, as they are necessary for many parts of smart devices (Crawford & Joler 2018, Walt 2018, Hammer & Park 2021). Data colonialism (Couldry & Mejias 2019) is accompanied by "data orientalism," with allegedly universal algorithms reproducing Western categories and constructing a distant other (Kotliar 2020).[5]

[5] Scholars often invoke a reference to (digital, data, algorithmic) colonialism by pointing at some similarities with colonial practices, often in regard to exploitation and appropriation. We believe that to preserve the

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

A final risk is that emergent digital technology sectors may, as in high-income countries, enhance rather than reduce in-country inequities. Poggiali (2017) describes how digital citizenship perpetuated class inequality between Nairobi's elite technology sector and its vast population of urban poor. Worse, new technologies rely on and enable new forms of exploiting the poor. Mobile phones allow for a new illicit industry of unregulated loans that put people in perpetual debt, for example (Donovan & Park 2019). While M-Pesa offers the unbanked access to services and cultivates even cash-strapped users through the introduction of per-second billing (Hughes & Lonie 2007, Park & Donovan 2016), it is criticized for unfair practices and inadequate data protection. Indeed, data insecurity is a major challenge for enterprises in low-ranked knowledge countries (Olatokun & Kebonye 2010).

Locals may be exploited not only as users but as workers. Even knowledge-based companies require low-skilled labor, at times employed under exploitative or even traumatizing conditions. One example is workers employed as content moderators, who filter social media platforms for posts that violate the platforms' rules (Roberts 2019); another example is workers for Amazon Mechanical Turk (Bergvall-Kåreborn & Howcroft 2014). The bulk of such labor is in the Global South (Gray & Suri 2017, Hammer & Park 2021). In turn, the high-skilled, knowledge-based jobs that exist in the Global South may be replaced in the future. Gaus & Hoxtell (2019) anticipate that in sub-Saharan Africa, automation will likely be at the expense of skilled workers in the high-wage and internationalized manufacturing sector and in its high-wage service economy.

To summarize, knowledge-based sectors exist also in lower-ranked knowledge economy countries, and we can learn much from including them in our analysis of knowledge production and use. The study of pockets of innovation in contexts of scarce knowledge resources offers new insights beyond those based on higher-ranked knowledge economies. It also offers a more critical perspective on the risks and costs of technology absorption, including dependence, inappropriateness, appropriation, exploitation, and exclusion.

4. STATE-LED KNOWLEDGE-BASED INNOVATION IN THE GLOBAL SOUTH: WELFARE AND SURVEILLANCE

We know from the development and the science and technology literatures that, often for military purposes, the state actively contributes to technological innovation. State innovations may be later appropriated by the private sector—as the invention of the Internet exemplifies (Kleine & Unwin 2009). The scholarship has paid less attention to innovations by and for state-internal functions, especially in the Global South. But much can be learned by looking at technologically rich state practices. The case of China is illuminating in this regard. China's goal of leading the world in knowledge-based development includes AI, but China is ahead only in the AI areas of facial recognition and smart surveillance—areas that do not "seem to have that many productivity spillovers to other industries" (O'Meara 2019, Ding 2020). These areas have obvious political implications for both policing and welfare (Mukhopadhyay et al. 2013). A number of follow-up questions emerge. Is state-led innovation conducive to private-sector innovation? Is innovation driven by political, military, or otherwise noneconomic incentives different than innovation motivated by profit? What role does technological innovation play in enabling state capacity—not only in regard to development but in regard to population management more generally (Veeraraghavan 2013)? To see

analytical specificity that colonialism deserves, scholars should more systematically clarify the differences between colonial practices on the one hand and racialized capitalist practices on the other, as well as trace the means through which colonial legacies get reproduced.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

whether and how the literature addressed such questions, we focus on state-led knowledge-based innovation in two essential realms of managing the population: welfare and surveillance.

Welfare benefits and surveillance have in common the need to see the subjects (Scott 1998). Breckenridge (2005) identifies a paradoxical link between these two fields by documenting how, in South Africa, a biometric system of fingerprinting initially used under apartheid was utilized after democratization for the distribution of social benefits. Indeed, technologies of identification are foundational for both functions and create an uncomfortable tension. The use of biometrics for the provision of social benefits in South Africa is only one example for how states in the Global South use innovative knowledge-based technology for welfare purposes. India's Aadhaar system, the world's largest biometric identity database, with 1.2 billion subscribers, has reportedly been successful in its efforts to streamline welfare provision to beneficiaries (Khera 2011, Banerjee 2016). In China, AI has been used for transforming major welfare services, including healthcare (Chun 2020) and education (Liu 2020). In turn, the prevalence of biometric registration and identification tools in high-income countries makes us underestimate the transformative aspects of that technology in the Global South—by bypassing administrative capacity of documentary registration, which requires a literate population, technology can be utilized to fill the administrative void of a weak state. By bypassing the need to teach citizens how to write, technology weakens the state's interest in strengthening one of the capabilities most fundamental for freedom (Sen 1999). Experts have also devised technological tools that address the challenges of identification using indirect yet intrusive sources of identification. For example, Blumenstock (2016) describes the use of a machine learning algorithm to infer an individual's socioeconomic status from their history of mobile phone use, as a means to improve antipoverty programs.

As welfare-related technology has expanded, the literature focused on two issues in particular: security and effectiveness. First, the amount of information available to a state in a biometric ID system raises serious privacy concerns, especially in political contexts that are unstable or have a history of corruption (Donovan 2015). Second, the issue of biometric failure or falsification due to network issues and the technology itself, coupled with structural issues like the largely rural welfare-dependent groups being far from enrollment centers or network connection, remains a source of anxiety and stress for poor beneficiaries whose participation in the program is mandatory (Singh 2021). In Malawi, for example, a new national ID system faced challenges enrolling beneficiaries due to the lack of physical infrastructure, including reliable access to electricity, in much of the country (Malik 2020). This points to a paradox of biometric welfare systems in the Global South, which is that in many low-income countries with highly innovative and technical biometric welfare systems, core structural issues are not being addressed with the same level of public investment or innovative approaches. Low-income countries that are fostering pockets of innovation around welfare distribution grapple with the same issues as high-income countries—security, centralized network connectivity, public support—while simultaneously facing the limits of their infrastructure.

This discussion again underscores that knowledge-based innovation should not be conceived as a next step in industrialization but rather as a separate experience for states in the twenty-first century. This discussion also points at the limits of the current orientation of the scholarly debate, which tends to focus more on questions regarding implementation and less on the innovation itself or the production of knowledge and social organization required for its success.

Identification is essential for redistribution but it is a tool of surveillance as well. State agencies may extract data using private-sector technology—Safaricom, the mobile network operator behind M-Pesa, was reportedly tasked with building a surveillance system for Kenya's national security apparatus (Park & Donovan 2016)—or states develop their own technologies (Chorev 2021). State agencies also use data that private technology companies collect on users (Bauman et al.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

2014, Amnesty Int. 2019, Zuboff 2019, Lindner & Xiao 2020). State surveillance tools may target minorities explicitly, as with the use of facial recognition technologies for the surveillance and detention of Muslim ethnic minorities in China's Xinjiang province, or the targeting of minorities could be due to algorithmic and other biases. Important work has been done on how formally neutral technologies used by state agencies responsible for both welfare and policing are inherently discriminatory, often on the basis of race, ethnicity, and/or class (Eubank 2018, Benjamin 2019). State surveillance may have other disciplinary and/or governmental functions. One of the most unique uses of digital technology is China's social credit system, which "takes aim at nearly all of China's development ills—from environmental protection to [intellectual property] and financial fraud to academic plagiarism" (Lewis 2020, p. 43). The social credit system is designed to encourage individuals, business, legal institutions, and government itself to be more "trustworthy" by scoring behavior and penalizing or rewarding based on one's score (Lewis 2020, p. 43; Liang et al. 2018; Zhang 2020). Although the literature has offered important insights on the use of state surveillance, we still know surprisingly little about the technological and political-economic conditions allowing for the development of such technologies. In addition, studies focus on local use rather than the political-economic conditions for and implications of exporting surveillance technology. China, Israel, and other countries export their military and policing technologies to others. Zimbabwe, Uzbekistan, Pakistan, Kenya, the United Arab Emirates, and Germany are using Chinese-made intelligent monitoring systems, for example. Even when the companies selling the technology are private, as Huawei is, the purchase is made possible for governments that could not otherwise afford it thanks to loans from the Chinese government (Mozur et al. 2019). In 2018, Citizen Lab confirmed the use of the Israeli-developed spyware Pegasus—often targeting members of the press and civil society—in 45 countries, including in sub-Saharan Africa, the Middle East and North Africa, the Gulf countries, Europe, and South and North America (Marczak et al. 2018). Although Pegasus is privately owned, the Israeli Defense Ministry must approve any license to a government that wants to buy it.

In turn, technology and technological innovation are used to resist state surveillance and other government practices. The use of social media for organizing and mobilizing protests is one important example (Tufekci & Wilson 2012). Another example, from Kenya, is an online platform called Ushahidi (testimony in Kiswahili) that, to pacify ethnic conflicts, aggregates information about events related to ethnic violence that users have witnessed (Poggiali 2017). One of the attractive features of Ushahidi is its appropriateness in low-technology settings, which makes it easy to transport to other places. Al Jazeera, for example, used the platform to monitor destruction during the 2008–2009 Gaza War. (Going back to the issue of return migrants, it is noteworthy that the developers were four Kenyan bloggers living in the diaspora.) In another piece, Poggiali (2016, p. 388) describes a different "techno-utopian digital mapping project," where Kenyan activists mapped the lack of social services in informal settlements so that they could not be ignored by the government. These examples demonstrate that ICT creates political capacities.[6]

When it comes to technological innovation by and for the state, we find that the existing literature happens in parallel to rather than in conversation with the literature on innovation in the private sector, with very different types of questions and concerns. Although the existing

[6]Aid agencies, in turn, seek initiatives that aim to employ ICT for enhancing individual capabilities (what is often referred to as ICT for development or ICT4D) (Toyama 2011). However, the rosy promises of ICT4D have been met with criticism, including by former proponents (Blumenstock 2018). Wade (2002) suggests that touting ICT tools as though they can leapfrog over development problems "is like saying that cheap books can cure illiteracy" (Wade 2002, p. 443). Worse, Toyama's (2011) amplifier theory suggests that the greater one's capacity, the more technology delivers; conversely, the lesser one's capacity, the less value technology has.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

discussions are illuminating, we believe that much can be gained by looking at the state as another site of innovation to better understand the political economy of innovation and potentially gain a different perspective on the state.

5. CONCLUSION

Current debates about knowledge-based economies in sociology remain largely limited to innovation and use in countries already endowed (and accordingly ranked) with relevant capabilities and proven accomplishments. However, as countries in the Global South push for development and use of new technologies, we must seek to understand the conditions under which knowledge-based economies, industries, and initiatives flourish or fail in contexts that have different historical trajectories.

Our review intentionally focuses on neglected areas of research. Bringing together literatures from economic sociology and the sociology of development, we first identified the main factors considered by scholars to be conducive for the use and/or production of knowledge, including in the Global South. Those focus on a number of domestic considerations, including state capabilities and support, even if scholars have come to see the main role of the state as supporting networks and creating links among domestic actors and between domestic actors and international knowledge carriers. Those conditions echo the conditions identified by the literature for development in general. In this review, however, we challenge the assumption that innovation is simply another step in industrial development, made possible by the presence of the same conducive factors. Instead, we call for a departure from this linear narrative. Additionally, we insist that understanding innovation in the Global South—and formerly colonized countries in particular—demands that sociologists consider the historically and geopolitically specific context of the countries and industries they encounter. To this end, we highlight instances of pockets of innovation in low- and middle-income countries. We show that despite legacies of historical marginalization and exploitation, pockets of innovation occur throughout the Global South. Instances of innovation are often rendered invisible in the literature on knowledge economies, but they add nuance to—and in some cases raise questions about—some of the insights that come out of looking at innovation in the more obvious places. We emphasize the risks and costs of technological diffusion and use, which may lead to technological dependence, inappropriate technological use, appropriation, and exploitation. In addition to pockets of innovation, we use the examples of welfare provision and surveillance to show the unique role of the state in innovation. We suggest that innovation developed and used by the state is likely to have different characteristics than innovation developed and used by the private sector, and we call for additional political-economic research in regard to both state actions and civil society's reactions.

Beyond expanding research on lower-ranked knowledge economies, pockets of innovation, and state action, there are many related avenues for future research on innovation that are promising and go against the grain of current research by paying attention to possible contradictions that come with the pursuit of growth. For example, there is a need for more research on transnational networks and relationships and their roles as sources of learning, investment, and innovation beyond market-led foreign direct investment—including government-to-government exchanges, return migration and foreign employees, and digital learning, as well as the geopolitical and cybersecurity dimensions of international competition. There is also a need for more research that brings together development and innovation with challenges related to climate change and inequalities, especially in countries with limited resources; and finally, there is a need to identify the historical legacies leading to the current uneven distribution of knowledge, as well as the potential costs and risks in innovation.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

LITERATURE CITED

Acemoglu D, Akcigit U. 2012. Intellectual property rights policy, competition and innovation. *J. Eur. Econ. Assoc.* 10(1):1–42

Akenzua K. 2019. TikTok: transforming video-sharing. *HBS Digital Initiative*, Oct. 18. **https://digital.hbs.edu/platform-digit/submission/tiktok-transforming-video-sharing/**

Altenburg T, Schmitz H, Stamm A. 2008. Breakthrough? China's and India's transition from production to innovation. *World Dev.* 36(2):325–44

Amnesty Int. 2019. *Surveillance giants: how the business model of Google and Facebook threatens human rights*. Rep., Amnesty Int., Mexico City

Amsden AH. 1989. *Asia's Next Giant: Late Industrialization in South Korea*. Oxford, UK: Oxford Univ. Press

Appelbaum R, Parker RA, Cao C, Gereffi G. 2011. China's (not so hidden) developmental state: becoming a leading nanotechnology innovator in the twenty-first century. In *State of Innovation: The US Government's Role in Technology Development*, ed. FL Block, MR Keller, pp. 217–35. London: Routledge

Archibugi D, Iammarino S. 2002. The globalization of technological innovation: definition and evidence. *Rev. Int. Political Econ.* 9(1):98–122

Aulakh PS. 2007. Emerging multinationals from developing economies: motivations, paths and performance. *J. Int. Manag.* 13(3):235–40

Banerjee S. 2016. *Aadhaar: digital inclusion and public services in India*. Rep., World Bank, Washington, DC

Bauman Z, Bigo D, Esteves P, Guild E, Jabri V, et al. 2014. After Snowden: rethinking the impact of surveillance. *Int. Political Sociol.* 8(2):121–44

Beck N. 1992. *Shifting Gears: Thriving in the New Economy*. Toronto: Harper Educ.

Beerepoot N, Keijser C. 2015. The service outsourcing sector as driver of development: the expectations of Ghana's ICT for Accelerated Development Programme. *Tijdschr. Econ. Soc. Geogr.* 106(5):556–69

Bell D. 1973. *The Coming of Post-Industrial Society: A Venture in Social Forecasting*. New York: Basic

Benjamin R. 2009. A lab of their own: genomic sovereignty as postcolonial science policy. *Politics Soc.* 28(4):341–55

Benjamin R. 2019. *Race After Technology: Abolitionist Tools for the New Jim Code*. Cambridge, UK: Polity

Bergvall-Kåreborn B, Howcroft D. 2014. Amazon Mechanical Turk and the commodification of labor. *New Technol. Work Employ.* 29(3):213–23

Block F. 2008. Swimming against the current: the rise of a hidden developmental state in the United States. *Politics Soc.* 36(2):169–206

Block F, Negoita M. 2016. Beyond embedded autonomy: conceptualizing the work of developmental states. In *The Asian Developmental State: Reexaminations and New Departures*, ed. Y Chu, pp. 57–72. London: Palgrave Macmillian

Blumenstock J. 2018. Comment: Don't forget people in the use of big data for development. *Nature* 561:170–72

Blumenstock JE. 2016. Fighting poverty with data. *Science* 353(6301):753–54

Bollou F, Ngwenyama O. 2008. Are ICT investments paying off in Africa? An analysis of total factor productivity in six West African countries from 1995 to 2002. *Inf. Technol. Dev.* 14(4):294–307

Bramann JU. 2016. *The Emergence of Kenya's "Silicon Savannah."* Diss. Thesis, HHL Leipzig Grad. Sch. Manag., Leipzig, Ger.

Breckenridge K. 2005. The biometric state: the promise and peril of digital government in the New South Africa. *J. S. Afr. Stud.* 31(2):267–82

Breznitz D, Murphree M. 2011. *Run of the Red Queen: Government, Innovation, Globalization, and Economic Growth in China*. New Haven, CT: Yale Univ. Press

Breznitz D, Ornston D, Samford S. 2018. Mission critical: the ends, means, and design of innovation agencies. *Ind. Corp. Change* 27(5):883–96

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

Burton-Jones A. 1999. *Knowledge Capitalism*. Oxford, UK: Oxford Univ. Press
Burton-Jones A. 2003. Knowledge capitalism: the new learning economy. *Policy Futures Educ.* 1(1):143–59
Cader HA. 2008. The evolution of the knowledge economy. *J. Reg. Anal. Policy* 38(2):117–29
Chang H-J. 2002. *Kicking Away the Ladder: Development Strategy in Historical Perspective*. London: Anthem
Chaudhuri S. 2005. *The WTO and India's Pharmaceuticals Industry: Patent Protection, TRIPS, and Developing Countries*. Oxford, UK: Oxford Univ. Press
Chen L. 2014. Varieties of global capital and the paradox of local upgrading in China. *Politics Soc.* 42(2):223–52
Chen L. 2017. Grounded globalization: foreign capital and local bureaucrats in China's economic transformation. *World Dev.* 98:381–99
Chibber V. 2003. *Locked in Place: State-Building and Late Industrialization in India*. Princeton, NJ: Princeton Univ. Press
Chorev N. 2007. *Remaking U.S. Trade Policy: From Protectionism to Globalization*. Ithaca, NY: Cornell Univ. Press
Chorev N. 2020. *Give and Take: Developmental Foreign Aid and the Pharmaceutical Industry in East Africa*. Princeton, NJ: Princeton Univ. Press
Chorev N. 2021. The virus and the vessel, or: how we learned to stop worrying and love surveillance. *Socio-Econ. Rev.* 19:1497–513
Christensen CM, Ojomo E, Dillon K. 2019. Cracking frontier markets. *Harvard Business Review*, Jan./Feb. **https://hbr.org/2019/01/cracking-frontier-markets**
Chun A. 2020. Breaking the iron triangle: AI in China's healthcare system. In *The AI Powered State: China's Approach to Public Sector Innovation*, ed. H Elliott, pp. 19–27. London: Nesta
Collinge C, Staines A. 2009. Rethinking the knowledge-based economy. *Built Environ.* 35(2):165–72
Comin D, Mestieri M. 2018. If technology has arrived everywhere, why has income diverged? *Am. Econ. J. Macroecon.* 10(3):137–78
Cooke P, Leydesdorff L. 2006. Regional development in the knowledge-based economy: the construction of advantage. *J. Technol. Transf.* 31:5–15
Couldry N, Mejias UA. 2019. *The Costs of Connection: How Data Is Colonizing Human Life and Appropriating It for Capitalism*. Stanford, CA: Stanford Univ. Press
Crawford K, Joler V. 2018. *Anatomy of an AI system: the Amazon Echo as an anatomical map of human labor, data and planetary resources*. Rep., AI Now Inst. Share Lab, New York Univ., New York
Davenport TH. 2005. *Thinking for a Living*. Cambridge, MA: Harvard Bus. Sch.
Davis J. 2019. The TikTok strategy: using AI platforms to take over the world. *INSEAD Knowledge*, June 19. **https://knowledge.insead.edu/entrepreneurship/the-tiktok-strategy-using-ai-platforms-to-take-over-the-world-11776**
Demena BA, van Bergeijk PAG. 2019. Observing FDI spillover transmission channels: evidence from firms in Uganda. *Third World Q.* 40(9):1708–29
Dent C. 2003. Transnational capital, the state and foreign economic policy: Singapore, South Korea, and Taiwan. *Rev. Int. Political Econ.* 10(2):246–77
DiMaggio P, ed. 2001. *The Twenty-First-Century Firm: Changing Economic Organization in International Perspective*. Princeton, NJ: Princeton Univ. Press
Ding J. 2018. *Deciphering China's AI dream: the context, components, capabilities, and consequences of China's strategy to lead in the world in AI*. Tech. Rep., Future Humanit. Inst., Oxford, UK
Ding J. 2020. Is China taking the lead in AI? *MIT Sloan Management Review*, April 30. **https://sloanreview.mit.edu/article/is-china-taking-the-lead-in-ai/**
Donovan KP. 2015. The biometric imaginary: bureaucratic techno-politics in post-apartheid welfare. *J. S. Afr. Stud.* 41:815–33
Donovan KP, Park E. 2019. Perpetual debt in the Silicon Savannah. *Boston Review*, Aug. 14. **https://bostonreview.net/articles/kevin-p-donovan-emma-park-tk/**
Drucker PF. 1993. *Post-Capitalist Society*. New York: Harper Bus.
Duysters G, Jacob J, Lemmens C, Jintian Y. 2009. Internationalization and technological catching up of emerging multinationals: a comparative case study of China's Haier group. *Ind. Corp. Change* 18(2):325–49
Eliasson G, Fölster S, Lindberg T, Pousette T, Taymaz E. 1991. *The Knowledge Based Information Economy*. Stockholm: Ind. Inst. Econ. Soc. Res.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

Eubank V. 2018. *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*. New York: St. Martin's

Evans P. 1995. *Embedded Autonomy: State and Industrial Transformation*. Princeton, NJ: Princeton Univ. Press

Ewusi-Mensah K. 2012. Problems of information technology diffusion in sub-Saharan Africa: the case of Ghana. *Inf. Technol. Dev.* 18(3):247–69

Ford LR. 2017. Intellectual property and industrialization: legalizing hope in economic growth. *Theory Soc.* 46:57–93

Fransen J, Helmsing AHJ. 2017. Absorptive capacity as a mediator: innovation of handicraft exporters in Yogyakarta, Indonesia. *Tijdschr. Econ. Soc. Geogr.* 108(6):737–51

Fuentes A. 2019. The divergent logics of industrial change: a comparison of export-cheese processors in Nicaragua. *Socio-Econ. Rev.* 19(1):333–58

Gaus A, Hoxtell W. 2019. *Automation and the future of work in sub-Saharan Africa*. Work. Pap., Glob. Public Policy Inst., Berlin

Gereffi G. 1978. Drug firms and dependency in Mexico: the case of the steroid hormone industry. *Int. Organ.* 32(1):237–86

Gereffi G, Ong R. 2006. *Upgrading in the global knowledge economy: insights from China and India*. Paper presented at Global Value Chains Workshop on Industrial Upgrading, Offshore Production, and Labor, November 9–10, Duke University, Durham, NC

Goldemberg J. 1998. Leapfrog energy technologies. *Energy Policy* 26(10):729–41

Gray ML, Suri S. 2017. The humans working behind the AI curtain. *Harvard Business Review*, Jan. 9. **https://hbr.org/2017/01/the-humans-working-behind-the-ai-curtain**

Guillén MF, García-Canal E. 2009. The American model of the multinational firm and the "new" multinationals. *Acad. Manag. Perspect.* 23(2):23–35

Halpern S. 2019. The terrifying potential of the 5G network. *New Yorker*, April 26. **https://www.newyorker.com/news/annals-of-communications/the-terrifying-potential-of-the-5g-network**

Hammer R, Park TM. 2021. The ghost in the algorithm: the veiling of racial colonial violence from slavery to the digital age. *Political Power Soc. Theory* 38:219–47

Harrison E, Mdee A. 2018. Entrepreneurs, investors and the state: the public and the private in sub-Saharan African irrigation development. *Third World Q.* 39(11):2126–41

Headrick DR. 1988. *The Tentacles of Progress: Technology Transfer in the Age of Imperialism, 1850–1940*. Oxford, UK: Oxford Univ. Press

Heidenreich M. 2012. The social embeddedness of multinational companies: a literature review. *Socio-Econ. Rev.* 10:549–79

Hicks J. 2019. 'Digital colonialism': why some countries want to take control of their people's data from big tech. *The Conversation*, Sept. 26. **https://theconversation.com/digital-colonialism-why-some-countries-want-to-take-control-of-their-peoples-data-from-big-tech-123048**

Hou X, Li P. 2020. Whose legitimacy? China's drive for electric vehicles. *Sociol. Dev.* 6(1):66–90

Hughes N, Lonie S. 2007. M-Pesa: mobile money for the "unbanked" turning cellphones into 24-hour tellers in Kenya. *Innovations* 1(2):63–81

IMF (Int. Monet. Fund). 2018. *Measuring the digital economy*. Policy Pap., IMF, Washington, DC

Jones C. 1999. Growth: with or without scale effects? *Am. Econ. Rev. Pap. Proc.* 89:139–44

Joo SH, Oh C, Lee K. 2016. Catch-up strategy of an emerging firm in an emerging country: analyzing the case of Huawei vs. Ericsson with patent data. *Int. J. Technol. Manag.* 72(1–3):19–42

Kaska K, Beckvard H, Minárik T. 2019. *Huawei, 5G and China as a security threat.* Work. Pap., NATO Coop. Cyber Def. Cent. Excell., Tallinn, Estonia

Khera R. 2011. The UID project and welfare schemes. *Econ. Political Wkly.* 46(9):38–43

Kim S-Y. 2012. Transitioning from fast-follower to innovator: the institutional foundations of the Korean telecommunications sector. *Rev. Int. Political Econ.* 19(1):140–68

Kim S-Y. 2019. Hybridized industrial ecosystems and the makings of a new developmental infrastructure in East Asia's green energy sector. *Rev. Int. Political Econ.* 26(1):158–82

Kleine D, Unwin T. 2009. Technological revolution, evolution and new dependencies: what's new about ICT4D? *Third World Q.* 30(5):1045–67

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

Kotliar DM. 2020. Data orientalism: on the algorithmic construction of the non-Western other. *Theory Soc.* 49:919–39

Krippner GR. 2012. *Capitalizing on Crisis: The Political Origins of the Rise of Finance*. Cambridge, MA: Harvard Univ. Press

Kwon H-K. 2005. National model under globalization: the Japanese model and its internationalization. *Politics Soc.* 33(2):234–52

Langbein J, Markiewicz O. 2019. Changing modes of market integration, domestic developmental capacities and state-business alliances: insights from Turkey's automotive industry. *Rev. Int. Political Econ.* 27:1104–25

Larraín F, Lopez-Calva LF, Rodriguez-Clare A. 2000. *Intel: a case study of foreign direct investment in Central America*. Work. Pap., Cent. Int. Dev., Harvard Univ., Cambridge, MA

Lee K. 2013. *Schumpeterian Analysis of Economic Catch-Up: Knowledge, Path-Creation, and the Middle-Income Trap*. Cambridge, UK: Cambridge Univ. Press

Lee K. 2019. *The Art of Economic Catch-Up: Barriers, Detours and Leapfrogging in Innovation Systems*. Cambridge, UK: Cambridge Univ. Press

Lee K, Lim C. 2001. Technological regimes, catching-up and leapfrogging: findings from the Korean industries. *Res. Policy* 30(3):459–83

Lee Y-S, Heo I, Kim H. 2014. The role of the state as an inter-scalar mediator in globalizing liquid crystal display industry development in South Korea. *Rev. Int. Political Econ.* 2(1):102–29

Lewis D. 2020. Separating myth from reality: how China's social credit system uses public data for social governance. In *The AI Powered State: China's Approach to Public Sector Innovation*, ed. H Elliott, pp. 43–50. London: Nesta

Li X. 2012. Behind the recent surge of Chinese patenting: an institutional view. *Res. Policy* 41(1):236–49

Liang F, Das V, Kostyuk N, Hussain MM. 2018. Constructing a data-driven society: China's social credit system as a state surveillance infrastructure. *Policy Internet* 10(4):415–53

Lindner AM, Xiao T. 2020. Subverting surveillance or accessing the dark web? Interest in the Tor anonymity network in U.S. states, 2006–2015. *Soc. Curr.* 7(4):352–70

Liu AM, Liang OX, Tuuli M, Chan I. 2018. Role of government funding in fostering collaboration between knowledge-based organizations: evidence from the solar PV industry in China. *Energy Explor. Exploit.* 36(3):509–34

Liu Y-L. 2020. The future of the classroom? China's experience of AI in education. In *The AI Powered State: China's Approach to Public Sector Innovation*, ed. H Elliott, pp. 27–34. London: Nesta

Lundvall BÅ, ed. 1992. *National Systems of Innovation: Towards a Theory of Innovation and Interactive Learning*. London: Pinter

Machlup F. 1962. *The Production and Distribution of Knowledge in the United States*. Princeton, NJ: Princeton Univ. Press

Maggor E. 2020. The politics of innovation policy: building Israel's "neo-developmental" state. *Politics Soc.* 49(4):451–87

Malik T. 2020. *Malawi's journey towards transformation: lessons from its national ID project*. Rep., Cent. Glob. Dev., Washington, DC

Marczak B, Scott-Railton J, McKune S, Razzak BA, Deibert R. 2018. *Hide and seek: tracking NSO Group's Pegasus spyware to operations in 45 countries*. Res. Rep., Citizen Lab, Univ. Toronto, Toronto, Can.

Marshall A. 1890. *Principles of Economics*. London: Macmillan

Marx K. 1847. *The Poverty of Philosophy*. Chicago: C.H. Kerr & Co.

Mathews JA. 2006. Dragon multinationals. *Asia-Pac. J. Manag.* 23:5–27

McDermott GA, Corredoira RA, Kruse G. 2009. Public-private institutions as catalysts of upgrading in emerging market societies. *Acad. Manag. J.* 52(6):1270–96

McGranahan DA, Beale CL. 2002. Understanding rural population loss. *Rural Am.* 17(Winter):1–10

Mehri DB. 2015. The role of engineering consultancies as network-centered actors to develop indigenous, technical capacity: the case of Iran's automotive industry. *Socio-Econ. Rev.* 13(4):747–60

Miller JL. 2016. *Nollywood Central: The Nigerian Videofilm Industry*. London: Br. Film Inst.

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

Mozur P, Kessel JM, Chan M. 2019. Made in China, exported to the world: the surveillance state. *New York Times*, April 24. **https://www.nytimes.com/2019/04/24/technology/ecuador-surveillance-cameras-police-government.html?auth=link-dismiss-google1tap**

Mu Q, Lee K. 2005. Knowledge diffusion, market segmentation and technological catch up: the case of the telecommunication industry in China. *Res. Policy* 34:759–83

Mukhopadhyay P, Muralidharan K, Niehaus P, Sukhtankar S. 2013. *Implementing a biometric payment system: the Andhra Pradesh experience*. Work. Pap., Univ. Calif., San Diego

Muzaka V. 2019. India and Brazil in pursuit of the competitive knowledge economy. *Rev. Int. Stud.* 45(2):345–64

Ndemo B, Weiss T, eds. 2017. *Digital Kenya: An Entrepreneurial Revolution in the Making*. London: Palgrave Macmillian

Nee V, Opper S. 2012. *Capitalism from Below: Markets and Institutional Change in China*. Cambridge, MA: Harvard Univ. Press

Negoita M, Block M. 2012. Networks and public policies in the Global South: the Chilean case and the future of the developmental network state. *Stud. Comp. Int. Dev.* 47:1–22

Ó Riain S. 2004. *The Politics of High-Tech Growth: Developmental Network States in the Global Economy*. Cambridge, UK: Cambridge Univ. Press

OECD (Organ. Econ. Co-op. Dev.). 1996. *The knowledge-based economy*. Work. Pap. GD(96)102, OECD, Paris

Olatokun W, Kebonye M. 2010. e-Commerce technology adoption by SMEs in Botswana. *Int. J. Emerg. Technol. Soc.* 8(1):42–56

O'Meara S. 2019. Will China overtake the U.S. in artificial intelligence research? *Scientific American*, Aug. 24. **https://www.scientificamerican.com/article/will-china-overtake-the-u-s-in-artificial-intelligence-research/**

Opertti F. 2018. Argentina is becoming a hub for knowledge-based services. *Beyond Borders Blog*, March 13. **https://blogs.iadb.org/integration-trade/en/argentina-is-becoming-a-hub-for-knowledge-based-services/#:~:text=Architectural%20plans%2C%20legal%20and%20accounting,(KBSs)%20that%20Argentina%20exports**

Park E, Donovan K. 2016. Between the nation and the state. *Limn* 7. **https://limn.it/articles/between-the-nation-and-the-state/**

Pérez C. 2001. Technological change and opportunities for development as a moving target. *CEPAL Rev.* 75:109–30

Perez-Aleman P, Alves FC. 2017. Reinventing industrial policy at the frontier: catalysing learning and innovation in Brazil. *Camb. J. Reg. Econ. Soc.* 10:151–71

Poggiali L. 2016. Seeing (from) digital peripheries: technology and transparency in Kenya's Silicon Savannah. *Cult. Anthropol.* 31(3):387–411

Poggiali L. 2017. Digital futures and analogue pasts? Citizenship and ethnicity in techno-utopian Kenya. *Africa* 87(2):253–77

Powell WW, Snellman K. 2004. The knowledge economy. *Annu. Rev. Sociol.* 30:199–220

Prasad M. 1998. International capital on "Silicon Plateau": work and control in India's computer industry. *Soc. Forces* 77(2):429–52

Pritchett L. 1997. Divergence, big time. *J. Econ. Perspect.* 11(3):3–17

Reslinger C. 2013. Is there an Asian model of technological emergence? *Socio-Econ. Rev.* 11(2):371–408

Rikap C. 2019. Asymmetric power of the core: technological cooperation and technological competition in the transnational innovation networks of big pharma. *Rev. Int. Political Econ.* 26(5):987–1021

Roberts ST. 2019. *Behind the Screen: Content Moderation in the Shadows of Social Media*. New Haven, CT: Yale Univ. Press

Rodrik D. 2016. Premature deindustrialization. *J. Econ. Growth* 21:1–33

Romer P. 1990. Endogenous technical change. *J. Political Econ.* 98:338–54

Romer P. 1993. Idea gaps and object gaps in economic development. *J. Monet. Econ.* 32:543–73

Rosenberg L, Brent A. 2020. Infrastructure disruption in "Silicon Savannah": exploring the idea of the creative class and their relation to quality of place in Nairobi, Kenya. *Int. J. Urban Reg. Res.* 44(5):809–920

Samford S. 2017. Networks, brokerage, and state-led technology diffusion in small industry. *Am. J. Sociol.* 122(5):1339–70

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

Saraswati J. 2008. The Indian IT industry and neoliberalism: the irony of a mythology. *Third World Q.* 29(6):1139–52

Saxenian A. 2006. *The New Argonauts: Regional Advantage in a Global Economy*. Cambridge, MA: Harvard Univ. Press

Saxenian A, Sabel C. 2008. Roepke Lecture in Economic Geography. Venture capital in the 'periphery': the new Argonauts, global search, and local institution building. *Econ. Geogr.* 84(4):379–94

Schumpeter JA. 1911. *The Theory of Economic Development: An Inquiry into Profits, Capital, Credit, Interest, and the Business Cycle*. Cambridge, MA: Harvard Univ. Press

Schumpeter JA. 1942. *Capitalism, Socialism and Democracy*. New York: Harper

Schüren V. 2013. What a difference a state makes: pharmaceutical innovation after the TRIPs agreement. *Bus. Politics* 15(2):217–43

Scott J. 1998. *Seeing like a State*. New Haven, CT: Yale Univ. Press

Sen A. 1999. *Development as Freedom*. New York, NY: Knopf

Shadlen KC. 2011. The political contradictions of incremental innovation: lessons from pharmaceutical patent examination in Brazil. *Politics Soc.* 39(2):143–74

Simons B. 2019. *Why "leapfrogging" in frontier markets isn't working*. Rep., Cent. Glob. Dev., Washington, DC

Singh P. 2021. Aadhaar and data privacy: biometric identification and anxieties of recognition in India. *Inf. Commun. Soc.* 24(7):978–93

Steinmueller E. 2002. Knowledge-based economies and information and communication technologies. *Int. Soc. Sci. J.* 171:141–53

Stewart F. 1977. *Technology and Underdevelopment*. London: Palgrave Macmillian

Takhteyev Y. 2012. *Coding Places: Software Practice in a South American City*. Cambridge, MA: MIT Press

Tang L, Murphree M, Breznitz D. 2016. Structured uncertainty: a pilot study on innovation in China's mobile phone handset industry. *J. Technol. Transf.* 41:1168–94

Tang M, Hussler C. 2011. Betting on indigenous innovation or relying on FDI: the Chinese strategy for catching-up. *Technol. Soc.* 33(1/2):23–35

Taniguchi R, Babb S. 2009. The global construction of development models: the US, Japan and the East Asian miracle. *Socio-Econ. Rev.* 7:277–303

Taylor MZ. 2012. Toward an international relations theory of national innovation rates. *Secur. Stud.* 21(1):113–52

Taylor MZ. 2016. *The Politics of Innovation: Why Some Countries Are Better than Others at Science and Technology*. Oxford, UK: Oxford Univ. Press

Thurbon E, Weiss L. 2019. Economic statecraft at the frontier: Korea's drive for intelligent robotics. *Rev. Int. Political Econ.* 28:103–27

Toyama K. 2011. Technology as amplifier in international development. In *Proceedings of the 2011 iConference*, pp. 75–82. New York: ACM

Tufekci Z, Wilson C. 2012. Social media and the decision to participate in political protest: observations from Tahrir Square. *J. Commun.* 62(2):363–79

Tyce M. 2020. Beyond the neoliberal-statist divide on the drivers of innovation: a political settlements reading of Kenya's M-Pesa success story. *World Dev.* 125:104621

Veeraraghavan R. 2013. Dealing with the digital panopticon: the use and subversion of ICT in an Indian bureaucracy. In *Proceedings of the 6th International Conference on Information and Communication Technologies and Development*, pp. 248–55. New York: ACM

Wade M, Shan J. 2019. TikTok: the world's most valuable startup that you've never heard of. *The Conversation*, Jan. 7. **https://theconversation.com/tiktok-the-worlds-most-valuable-startup-that-youve-never-heard-of-109302**

Wade RH. 2002. Bridging the digital divide: new route to development or new form of dependency? *Glob. Gov.* 8(4):443–66

Walt V. 2018. Blood, sweat, and batteries. *Fortune*, Aug. 23. **https://fortune.com/longform/blood-sweat-and-batteries/**

Weiss L. 2005. Global governance, national strategies: how industrialized states make room to move under the WTO. *Rev. Int. Political Econ.* 12(5):723–49

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08

Whittaker DH, Zhu T, Sturgeon T, Tsai MH, Okita T. 2010. Compressed development. *Stud. Comp. Int. Dev.* 45(4):439–67

Wojan T. 2000. The composition of rural employment growth in the "new economy." *Am. J. Agric. Econ.* 82(Aug.):594–605

Wong J. 2011. *Betting on Biotech: Innovation and the Limits of Asia's Developmental State*. Ithaca, NY: Cornell Univ. Press

World Bank. 2007. *Building knowledge economies: advanced strategies for development*. Rep., World Bank, Washington, DC

World Bank. 2019. *Innovative China: New Drivers of Growth*. Washington, DC: World Bank

Yeung HW. 2014. Governing the market in a globalizing era: developmental states, global production networks and inter-firm dynamics in East Asia. *Rev. Int. Political Econ.* 21(1):70–101

Zhang C. 2020. Governing (through) trustworthiness: technologies of power and subjectification in China's social credit system. *Crit. Asian Stud.* 52(4):565–88

Zhang Y, Duysters G. 2009. Alliance-based network view on Chinese firms' catching-up: case study of Huawei Technologies Co. Ltd. *RISUS* 1(2). **https://doi.org/10.24212/2179-3565.2010v1i2a9**

Zuboff S. 2019. *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. London: Profile

Downloaded from www.annualreviews.org. Guest (guest) IP: 170.244.3.166 On: Tue, 15 Sept 2026 16:08:08